Do World Action Models Act on What They Imagine? Causal Evidence from Patching the Imagined Future

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***Abstract*—World action models (WAMs) generate a short video of the future and then predict robot actions from it. Whether the imagined future matters at test time is disputed, and the debate rests on task success rates alone. We open the model instead. Using Fast-WAM, whose released checkpoint acts from one set of weights through a per-layer key/value cache of its imagined video, we replace the imagined future inside that cache and measure the effect on the action and on closed-loop behavior in LIBERO. Replacing the imagined future with the one imagined for another task turns the predicted action into that task's action (a change 40× the sampling noise floor). In closed loop, a robot told to do task A while imagining task B completes B in 113 of 150 episodes (75%) and A in 4; telling it B while it imagines A leaves it doing A (94%). The effect is localized: patching layers 21–30 alone suppresses task A, while patching layers 1–20 changes nothing, even though attention to the future peaks in the middle layers. An action lens shows why: early layers do encode the imagined task, but the late layers decide, and they overwrite earlier layers when the two disagree. The action reads mainly the tokens where the imagined futures differ, through a few attention heads. The imagined future is therefore a strong control channel of this WAM, with direct consequences for monitoring and steering such policies.**

***Index Terms*—world action models, mechanistic interpretability, activation patching, robot manipulation, video generation**

I. Introduction

Robot policies built on pretrained video generators, which we call world action models (WAMs) following recent usage [1], [2], predict future video and actions together [1], [3]–[6]. A common design first *imagines* the next frames and then infers the actions that would produce them, in the manner of an inverse dynamics model (IDM). Imagining is expensive: it multiplies inference time. This raises a basic question with a practical cost attached: *does the policy act on what it imagines?*

The question is open. Fast-WAM [2] reports that removing test-time imagination costs little, while removing video co-training costs much, and concludes that the benefit of video lies in training. A follow-up disputes the test-time claim [7]. Both arguments compare success rates between models or modes. A success rate cannot tell whether a policy that imagines actually *reads* its imagination: a model could imagine faithfully and act from the current frame alone, or lean on the imagined future and still succeed for other reasons.

We answer the question causally, from inside the model. Fast-WAM's released Optional-IDM checkpoint is well suited to this. One set of weights imagines a future clip with a video transformer, freezes it into a per-layer key/value (KV) cache, and an action transformer reads that cache through shared attention. The imagined future thus sits in a known set of cache positions at every layer, where it can be read, removed, replaced or blended without retraining. We use this handle in three stages, each building on the last:

1) **Open loop.** We validate an exact re-implementation of the inference pipeline and replace the imagined future with alternatives. The action follows the imagined future, far more than the current frame, and the effect sits in the last ten layers although attention to the future peaks earlier (Sec. IV).
2) **Closed loop.** In simulation, swapping in another task's imagined future at every decision makes the robot perform the other task; swapping the instruction does not. Late layers alone are enough (Sec. V).
3) **Mechanism.** We locate what is read and where the decision forms: the tokens where imagined futures differ, decoded by early layers but decided by late layers, through a few heads (Sec. VI).

Recent work has shown that WAMs depend on their future cache: masking or scrambling it lowers closed-loop success [8], and attention to future frames has been proposed as a measure of reliance on them [9]. We go further and ask *what* the action takes from the imagined future and *where*. A coherent future imagined for another task redirects the robot to that task, while another task's instruction does not; the effect is carried by the last ten layers rather than by the layers that attend most to the future; and a lens on the action stream shows the decision forming there. To our knowledge this is the first layer-level causal account of how a WAM uses its imagined future, measured both on the action and on closed-loop task outcome.

II. Related Work

Video-based policies and WAMs. UniPi [3] plans by generating video and recovers actions with an inverse dynamics model; GR-1 [4] and Seer [5] predict future frames jointly with actions; recent WAMs build on large video

diffusion transformers [10], [11] and report strong zero-shot and dexterous performance [1], [6]. Behavioral comparisons find WAMs more robust than vision-language-action policies under perturbations [12]. Fast-WAM [2] and LAWA [7] disagree on whether test-time imagination is needed; both rely on success rates. Efficiency work caches or reuses the future key/value states that the action reads [8], [13]; RIFT in particular masks, shuffles or replaces the future cache of four WAMs in closed loop and finds that success drops [8]. These interventions establish that the future is used but not what is read from it, and they do not localize the effect in the network.

Interpretability of robot policies. Hong et al. [14] probe WAM activations for nuisance factors (camera pose, initial gripper state, image noise) and steer activations to recover success. They study whether a perturbation is linearly visible and steerable, pooling over all tokens; they do not separate the current frame from the imagined future or test what the action causally reads. Jiang et al. [15] find that in VLAs and in two WAMs, including Fast-WAM, an instructed target remains decodable from the policy's states yet does not govern the action; our instruction swap gives a causal counterpart in which the imagined future, not the instruction, governs it. Activation patching in VLAs locates the language and image pathways in a band of mid-to-late cross-attention layers [16], [17]. Mishra et al. [9] relate the action head's attention to future frames to generalization. Our question is complementary to all of these: which information from the imagined future the action uses, where, and with what behavioral consequence.

Causal interpretability methods. Activation patching, also called causal mediation or causal tracing, replaces internal states between runs to locate the components that carry a behavior [18], [19]; we follow the metric and baseline recommendations of [20], [21]. Attention weights alone are an unreliable account of what a model uses [22], [23]; we measure both and find them dissociated. Lenses decode intermediate states with the output layer, raw [24] or with a learned per-layer correction [25]. A raw action lens has been applied to robot policies [15]; we use a tuned lens on patched runs to follow where the decision changes.

III. Model and Method

A. Fast-WAM

Fast-WAM [2] pairs the Wan2.2-TI2V-5B video diffusion transformer [10] (30 layers, width 3072) with a 1B-parameter action transformer (30 layers, width 1024) in a mixture-of-transformers design [26]: the two networks have separate weights, but at every layer they share one attention operation (24 heads of dimension 128). The released Optional-IDM checkpoint for LIBERO [27] runs in two modes from the same weights. In *IDM mode*, the video transformer generates a 9-frame clip by 10 flow-matching steps [28], [29], with the current frame clamped; the finished clip is passed through the video transformer once more to fill a KV cache at every layer; the action transformer then denoises a chunk of 32 seven-dimensional actions (end-effector translation and rotation, gripper) in 10 steps, its layer ℓ attending to cache layer ℓ and to the other action tokens. In *first-frame mode*, the action reads only the current frame.

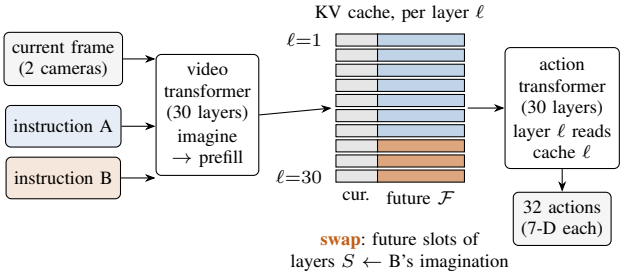


Fig. 1. Fast-WAM's IDM path and our intervention. The video transformer imagines a future clip and is run once more on the finished clip to fill a key/value (KV) cache at every layer: 98 current-frame tokens and 196 imagined-future tokens $\mathcal{F}$. Action layer ℓ attends to cache layer ℓ. We overwrite the future slots of chosen layers S with those computed from task B's imagination (here S = layers 21–30, orange), keeping the current frame and the instruction.

Both cameras (224×224 each) are placed side by side and encoded by the video autoencoder, which compresses 16× in space and 4× in time; with 2×2 patches, each latent frame becomes 98 tokens (a 7×14 grid, main camera on the left half). The cache therefore holds 294 tokens per layer: 98 for the current frame and 196, denoted $\mathcal{F}$, for two imagined latent frames. The video attention mask is causal by frame, so current-frame tokens never see the imagined future; their cache entries do not depend on it. The instruction reaches the action transformer separately, through cross-attention to text embeddings to which the robot's proprioceptive state is appended.

B. Patching the imagined future

Let $K_\ell^A, V_\ell^A \in \mathbb{R}^{294\times 3072}$ be the cache of a run imagined with instruction A, and K_ℓ^B, V_ℓ^B the cache computed from a *donor* imagination. For a set of layers S and a mixing weight α, we set

$$\tilde{K}_\ell[\mathcal{F}] = (1-\alpha)\, K_\ell^A[\mathcal{F}] + \alpha\, K_\ell^B[\mathcal{F}], \quad \ell \in S, \qquad (1)$$

and likewise for V, leaving all other entries, the instruction and the action noise unchanged. A swap is $\alpha = 1$. We also replace $\mathcal{F}$ with uninformative futures (the mean future over states, Gaussian noise matched in mean and variance, or the cache of a static clip that repeats the current frame), patch the current-frame slots instead, or mask attention from actions to $\mathcal{F}$ entirely (*knockout*). Finer variants restrict the swap to one imagined frame, one camera, a subset of tokens, or the 128 channels of a single head.

Measures. Actions are compared in the model's normalized action space (each dimension min-max scaled to $[-1, 1]$). The *action change* is $\Delta = \|\hat{a}' - \hat{a}\|_2$ over all 32×7 values between the edited and unedited action chunks, with identical noise. The *noise floor* is Δ obtained by resampling only the action noise. When the donor comes from task B with its own

unedited action $\hat{a}_B$, the *pull* toward B is $1-\|\hat{a}'-\hat{a}_B\|/\|\hat{a}-\hat{a}_B\|$ (0: no movement toward B; 1: B's action).

Action lens. With flow matching, the action head predicts a velocity v from which the clean action is estimated as $\hat{x}_0 = a_k - \sigma_k v$ at denoising step k with noise level σ_k. Since the head is a single linear map, we apply it to the action stream h_ℓ after each layer, $\hat{x}_0^{(\ell)} = a_k - \sigma_k W h_\ell$ (raw lens [24]), or first map h_ℓ to the last layer with a per-layer affine transform fitted by ridge regression on unedited runs (tuned lens [25]). The decoded action is scored as $d_A/(d_A + d_B)$, its relative distance to the unedited actions for A and for B (0: A, 1: B). We read the lens at the first denoising step, which starts from the same noise in every run, so differences reflect only the layer's state.

C. Protocol

Validation. We split the official inference function into imagine, prefill and act stages so that the cache can be edited. Run unedited, the split pipeline reproduces the official action bit for bit, a self-patch changes nothing, all checkpoint tensors load without missing keys, and the lens applied to the final layer reproduces the model's output exactly. Experiments use bf16 on single NVIDIA L40S GPUs (one decision: 0.31 s imagining, 0.28 s acting).

Tasks and states. LIBERO-Goal has ten tasks in one kitchen scene (drawers, stove, bowl, plate, wine bottle and rack, cream cheese), so the goal of any task can be checked on any simulator state. We pair task A with B $= (A+5) \bmod 10$ (e.g., "open the middle drawer of the cabinet" with "push the plate to the front of the stove"). Open-loop analyses use 100 states: first training demonstrations (10 tasks $\times$ 2 episodes $\times$ 5 time points, three seeds), then 100 fresh states saved from our own closed-loop rollouts (2 decisions from each of 50 episodes), which the model never trained on; the donor is a state of a different task. Closed-loop runs follow the official evaluation (wait 30 steps, re-plan every 10 steps, 400-step limit) for 10 tasks $\times$ 5 LIBERO starting states, at seed 42 and, for the main comparison, seeds 0 and 1. At every decision the donor is imagined from the robot's current observation with B's instruction, and B's goal is checked on the same simulator state. Intervals are 95% bootstrap over episodes (open loop) or Wilson intervals (success rates).

IV. Open Loop: The Action Follows the Imagined Future

The action is rewritten by the imagined future. Replacing the imagined future with one imagined for another task changes the action by 8.02 on fresh states, 40$\times$ the noise floor, and moves it to within 1.41 of that task's own action, from 8.68 (Table I). Replacing the current frame instead changes it by only 1.05. Uninformative futures (mean, static, noise) disturb the action by a fifth to a half as much without steering it anywhere. Conversely, on demonstrations where the true future is known, supplying it brings the action to 0.33 from the demonstrated action, against 0.88 with the model's own imagination: the action pathway behaves as an inverse dynamics reader of whatever future it is given. Two further observations bear on the test-time debate. Knocking out attention to the future reproduces first-frame mode exactly, confirming that the two modes differ only in this read. And on demonstrations, first-frame actions are as close to the demonstrated action as imagine-mode actions (0.85 vs. 0.88), so the imagination is not needed for open-loop accuracy, even though it dominates the action when present.

TABLE I
Action change when the imagined future (or the current frame) is replaced in all 30 layers. Training: 100 demonstration states, three seeds. Fresh: 100 states from our rollouts. Noise floor: 0.21 / 0.20.

Replacement	Training	Fresh
Future from another task	**7.64**	**8.02**
distance to that task's own action	2.05 (was 8.71)	1.41 (was 8.68)
Mean future	2.92	3.87
Static future (current frame repeated)	1.83	2.85
Noise future	1.57	2.40
Knockout (= first-frame mode)	1.02	2.04
Current frame from another task	1.38	1.05
Other task's future, layers 1–10	0.19	0.17
layers 11–20	0.66	0.66
layers 21–30	5.97	7.01

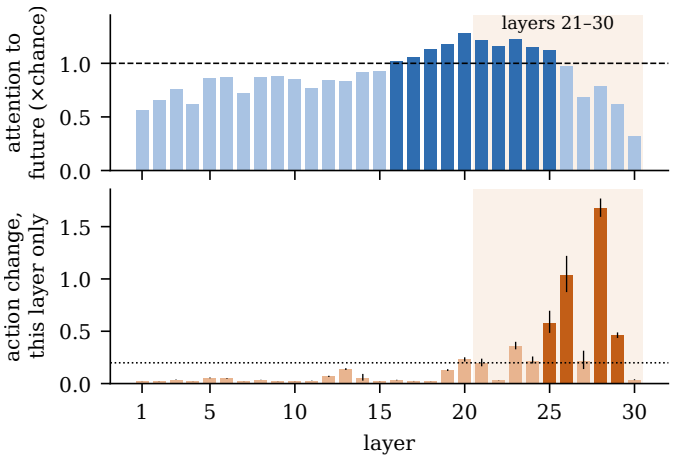


Fig. 2. Attention is not use (fresh states). Top: action-to-future attention per layer, relative to the future's share of tokens. Bottom: action change when the future is swapped in that layer only; dotted line: noise floor. Attention peaks around layer 20; the effect sits in layers 25–29.

The effect lives in the last ten layers, not where attention peaks. Swapping the future in layers 1–10 has no effect (0.17, below the floor), layers 11–20 a small one (0.66), and layers 21–30 almost the full effect (7.01). Swapping single layers locates the largest contributions at layer 28 (1.68) and 26 (1.04); single layers sum to 5.87, short of the all-layer 8.02, so layers act jointly (Fig. 2). We recorded the attention weights in the same runs, recomputing them in full precision (the grouped outputs reconstruct the model's attention output to within 5×10^{-3}). Action tokens give 19–77% of their attention to the future in every layer, and more than its token share only in layers 16–25 (peak 1.29$\times$ at layer 20); layers 27–30 attend

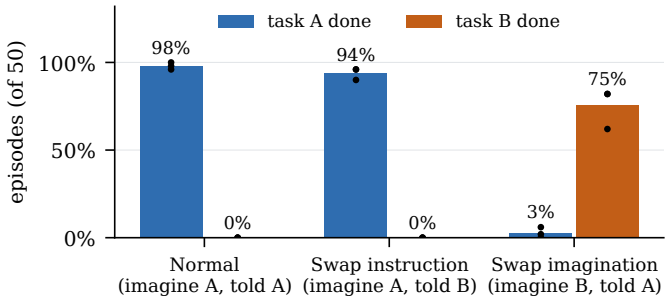


Fig. 3. Closed loop, 10 tasks × 5 starting states × 3 seeds. Bars: mean over seeds; dots: individual seeds. "Done" means the task's goal was satisfied in the simulator.

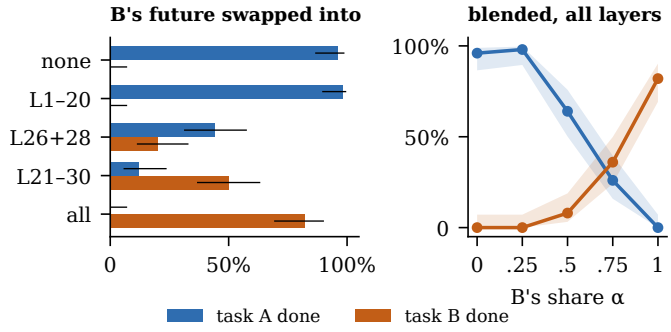


Fig. 4. Closed loop, seed 42, 50 episodes per row or point. Left: B's future swapped into some layers only; the rest keep A's. Right: futures blended in all layers, Eq. (1). Error bars and bands: 95% Wilson intervals.

mostly to the other action tokens (0.41–0.54). The layers that attend most to the future are thus not the ones whose future the action depends on, a dissociation of the kind cautioned against in [22]. All results replicate between training and fresh states.

V. Closed Loop: The Robot Does What It Imagines

We now apply Eq. (1) at every decision of a full episode. In the normal policy the robot completes task A in 147 of 150 episodes. If the action transformer is told B while the cache holds A's imagination, it still completes A in 141 of 150 episodes (48/45/48 per seed) and never B. If instead the cache holds B's imagined future while the action is told A, the robot completes A in 4 episodes and B in 113 (41/41/31 per seed; Fig. 3). It does so fluently: B's goal is reached after a median of 110 steps, against 90 for A in normal runs. B's goal check never fired in normal or instruction-swapped runs (0/300). The robot does not merely fail task A; it performs the task it imagines. The instruction still matters, through the imagination it produces, but given a fixed imagined future the action transformer's own access to the instruction has no measurable effect.

Late layers are sufficient; early layers are not. Restricting the swap to layers 1–20 leaves behavior intact (A 98%, B 0%). Restricting it to layers 21–30 suppresses task A (12%) and produces B in 50% of episodes; layers 26 and 28 alone, the two strongest single layers, do part of this (A 44%, B 20%; Fig. 4, left). Episodes that end in neither goal show mixed

TABLE II
Partial swaps of another task's imagined future (all layers, 100 fresh states). Pull: fraction of the way from the unedited action to the other task's action.

Swapped part of the future	Change	Pull
All 196 tokens	8.02	0.83
50% of tokens that differ most	**7.31**	**0.72**
Random 50% of tokens	3.25	0.24
50% of tokens that differ least	0.84	0.06
Imagined frame 1 / frame 2	4.63 / 5.37	0.25 / 0.31
Main camera / wrist camera	3.11 / 5.33	0.23 / 0.43

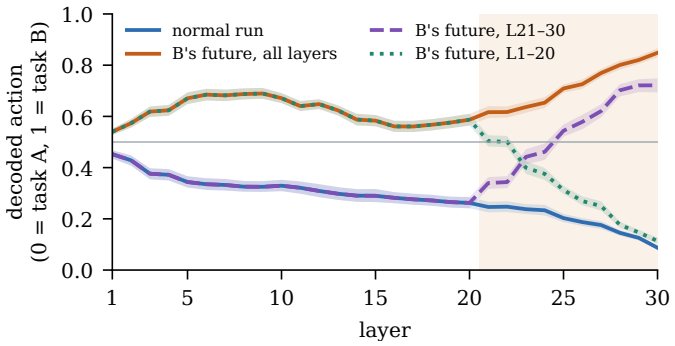


Fig. 5. Tuned action lens on 50 held-out fresh states (the lens was fitted on the other 50). Each curve: the action decoded after each layer, scored between task A (0) and task B (1); bands: 95% intervals over states.

behavior; for example, with the late layers swapped, a robot told to put the bowl on top of the cabinet while imagining putting the wine bottle on the rack picked up the bowl, set it down on the stove instead, and stayed there until the time limit.

A threshold, not a linear dial. Blending the two futures (Fig. 4, right) leaves behavior unchanged at $\alpha = 0.25$ (A 98%), mixed at 0.5 (A 64%, B 8%), and mostly flipped at 0.75 (A 26%, B 36%). A minority share of a foreign future is ignored; the policy commits once the foreign future dominates.

VI. Mechanism: What Is Read, and Where It Is Decided

What is read: where the futures differ. We rank the 196 future tokens by how much the two imagined latent videos differ at each position and swap a fraction of them (Table II). The half that differs most carries almost the full flip (pull 0.72 vs. 0.83); the half that differs least has nearly no effect (0.84), and a random half falls in between. The action thus reads the content that distinguishes one imagined outcome from another, such as the moving arm and objects, rather than the static scene. Neither imagined frame alone and neither camera alone suffices, and the wrist camera carries more than the main camera (pull 0.43 vs. 0.23).

Where it is decided: early layers read, late layers decide. The tuned action lens (Fig. 5) shows that when every layer sees B's future, the decoded action already leans toward B by layer 9 (0.69). This explains the strong early attention: information about the imagined future does enter early. Yet when only

layers 1–20 see B's future, the decoded action leans toward B up to layer 20 (0.59), exactly as in the full swap, and is then pulled back to A by layers 21–30, which see A's future (0.11 at the output, against 0.09 for the unedited run). When only layers 21–30 see B's future, the reverse happens: the action decodes as A until layer 20 (0.26) and moves to B afterward (0.72). The late layers therefore form the decision and override what earlier layers concluded, which reconciles the attention profile with the patching and closed-loop results. The raw lens, without a learned correction, shows the same late transition but no early signal, so the early reading is present but not in the output basis.

Which heads. Swapping the future in a single head of layer 26 or 28 identifies a few heads that carry most of each layer's effect (layer 26: h10 (0.45), h8 (0.29), h6 (0.15); layer 28: h23 (0.65), h15 (0.50), h12 (0.29)). No single head moves the action more than 0.04 of the way toward B, and a head's attention to the future predicts its effect only moderately (correlation +0.46 and +0.58). The read is thus distributed over a handful of heads in several late layers rather than concentrated in one "imagination head".

VII. Discussion

Imagination as a control channel. In this WAM, what the policy imagines determines what it does, more than what it is told. This has two sides. It is a risk: anything that corrupts the imagined future, whether a generation failure, a distribution shift in the video model, or an adversarial edit, redirects the robot to a coherent but wrong task, and the instruction does not act as a safeguard. It is also an opportunity: the imagined video is a readable forecast of the policy's next behavior, which can be inspected before execution, and the late layers and few heads we identify are natural targets for monitoring and for steering methods such as [14]. Adversarial attacks have been shown to decouple a WAM's actions from its imagination [30] or to corrupt the imagination of imagine-then-act models [31]; our results show that when the imagination is coherent but wrong, the action follows it.

On the test-time debate. Our results are compatible with Fast-WAM's finding that imagination is not required for success [2]: in open loop, first-frame actions are as accurate as imagine-mode actions. They add that when imagination is present, it is not ignored; it dominates the action. Success rates alone cannot distinguish these situations. How much a WAM depends on its future also varies with design: masking the future collapses success in a jointly denoising Fast-WAM variant [8], whereas in the Optional-IDM checkpoint masking reproduces the first-frame mode it was also trained for. Whether imagination helps under harder or perturbed conditions, the claim of [7], is a question our patching protocol can address directly.

Limitations. We study one WAM whose action transformer is trained as an inverse dynamics reader, so reliance on the future is encouraged by design; that it nonetheless ignores its own instruction access, and how and where the future is read, are not implied by the design, but other WAM architectures, in which video and actions are denoised jointly, need the same test. Our closed-loop evidence covers one LIBERO suite with ten tasks in one scene, five starting states per task and three seeds, in simulation; real-robot validation is future work. The B goal is checked every ten steps. The lens shows what is decodable, not what is used; the claim that late layers decide rests on the patching and closed-loop experiments, which agree with it.

VIII. Conclusion

We asked whether a world action model acts on what it imagines and answered by intervening on the imagined future inside the model. In Fast-WAM, it does: the imagined future rewrites the action, and in closed loop the robot carries out the task it imagines rather than the task it is told. The information enters early but is decided in the last ten layers, read from the tokens where imagined outcomes differ, through a few heads. Patching the imagined future gives a direct, causal tool for evaluating, monitoring and steering video-based robot policies.

Statements

Data and code availability. Code, per-episode results, analysis scripts and videos are available at https://github.com/TusharBajaj07/wam-imagination-patching. The model checkpoint and LIBERO data are public [2], [27]. **Author contributions.** T.B.: conceptualization, methodology, investigation, analysis, writing. **Use of AI tools.** Experiment code, analysis and an initial draft of this paper were prepared with assistance from an AI system (Claude, Anthropic). The author reviewed all code, results and text and takes full responsibility for them. **Funding and competing interests.** This work received no external funding. The author declares no competing interests.

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